

Signals & Systems (EE504) **Pre-Reading Material – Lecture 5**

Basic Concepts of LTI Systems and Convolution
Estimated Reading Time: 20–25 minutes

Course Context

Building on the system properties discussed in Lecture 4, this lecture introduces Linear Time-Invariant (LTI) systems, which are fundamental to the analysis of signals and systems. The core focus is the concept of convolution, a process that allows us to determine the output of a system for any arbitrary input by understanding its response to a single elementary signal: the unit impulse.

Why This Topic Matters

Most practical engineering systems can be approximated as LTI systems. The beauty of LTI systems lies in the principle of superposition; if we can decompose a complex input signal into simpler elementary components (impulses), we can find the total output by simply summing the system's responses to those individual components. This simplifies the analysis of everything from audio processing to communication networks.

Learning Outcomes

After completing this lecture, you should be able to:

- Define the characteristics of Linear Time-Invariant (LTI) systems.
- Explain the significance of the unit impulse response $h(t)$ or $h[n]$.
- Perform the convolution summation for discrete-time signals.
- Apply the properties of convolution, including commutativity, associativity, and distributivity.
- Determine the length of a convolved signal based on the lengths of the input signals.



Activate Your Prior Knowledge

Recall from previous lectures:

- A linear system must obey both the principles of superposition and homogeneity.
- A time-invariant system is one where a delay in the input leads to an identical delay in the output.
- The unit impulse function $\delta[n]$ or $\delta(t)$ is a signal that exists only at a single point in time with unit area/magnitude.

Prerequisite Concepts

- Signal Decomposition: The process of representing an arbitrary signal as a weighted sum of shifted unit impulses.
- Unit Impulse Response ($h[n]$): The specific output produced by a system when the input is a unit impulse $\delta[n]$.

Core Concepts

1. The Power of LTI Systems

LTI systems are uniquely simple to analyze because an arbitrary input $x[n]$ can be decomposed into a summation of scaled and shifted impulses. If the system is linear, we can find the response to each impulse individually; if it is time-invariant, we know that a shifted impulse will produce a similarly shifted version of the impulse response.

2. The Convolution Summation

For a discrete-time LTI system, the output $y[n]$ for any input $x[n]$ is given by the convolution summation:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

This formula shows that the output is a weighted sum of shifted impulse responses.

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Course Repository: <https://github.com/rajesh101191/Signals-Systems-PIEMR-2026-EE504>

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3. Properties of Convolution

Commutative

The order of signals does not matter:

$$x[n] * h[n] = h[n] * x[n]$$

Associative

In a cascade of systems, the order of processing does not change the final output:

$$x[n] * (h_1[n] * h_2[n])$$

Distributive

Convolution over a sum of impulse responses is equal to the sum of individual convolutions:

$$x[n] * h_1[n] + x[n] * h_2[n]$$

This represents parallel systems.

4. The Mechanism of Convolution

To visualize convolution physically, one follows four steps:

1. Flip: Reverse the impulse response $h[k]$ to get $h[-k]$.
2. Shift: Move the flipped signal by n to get $h[n-k]$.
3. Multiply: Multiply $x[k]$ and $h[n-k]$ sample by sample.
4. Sum: Add all the products to find the output at index n .

Guided Reading Questions

1. Why is it sufficient to know only the unit impulse response to fully characterize an LTI system?
2. In the context of the "honesty" analogy, how many counterexamples are needed to prove a system is not linear or not time-invariant?
3. If an input signal has a length of N_1 and an impulse response has a length of N_2 , what will be the total number of non-zero samples in the resulting output?
4. How does the time-invariance property simplify the calculation of responses to shifted impulses?

